paper_results: rebuilding the tables and figures

What this folder shows

This folder rebuilds every table and figure of the paper and compares each number with the printed one.

Result	What it contains	Script
Table 7	BI for a suite of models on eight sets of traces	`table7_bi_suite.py`
Table 8	how often the best model of each suite finds the key byte	`table8_key_recovery.py`
Table 9	BI for a suite of 80 models, widened to nearby models	`table9_biloc_full_suite.py`
Figure 1	BI stays stable while other information measures drift, as traces get longer	`figure1_dimension_sweep.py`
Figure 2	how far a model can be changed before its BI changes, and the chosen safety margin	`figure2_biloc_selected_centers.py`
full-version tables	two results quoted from the long version of the paper	`fullversion_tables.py`

`archived/` holds the saved counts from the runs behind the paper. `expected/` holds the printed numbers. `reproduce.py` runs the levels.

Before you start

Level	You need
A	nothing
B	the TCHES20 download (`fetch_tches20.sh`)
C	the Transformer source code, the model archive, `ASCAD.h5`, and `CHES_Challenge.h5`
D	the TCHES20 download; the raw ASCAD traces are fetched by the level itself

Level A: check the saved counts (half a minute, no data)

```
python -m paper_results.reproduce archived
```

```
$ python -m paper_results.reproduce archived
[... earlier output omitted ...]
[PASS] Figure 2 (selected-center local-family certificates): 12 values match the paper, 0 differ
[paper_results.figure2_biloc_selected_centers: 2 s, exit code 0]

$ python -m paper_results.fullversion_tables --recomputed-dir results/paper_results/none --out-dir results/p
    aper_results
### Full version: raw-ASCAD four-byte composition

| Traces | Printed | Printed rate | Archived final models (1,000 campaigns) | Rerun | Rerun rate | Rerun med
    ian rank |
|---|---|---|---|---|---|---|
```

```
| 100 | 40/250 | 0.160 | 0.203 |  |  |  |
| 200 | 94/250 | 0.376 | 0.445 |  |  |  |
| 500 | 191/250 | 0.764 | 0.918 |  |  |  |
[... 11 lines omitted ...]
[PASS] Four-byte composition (full version): 1 values match the paper, 0 differ
| ASCAD d100 | 22.48 | 9.43e-11 | yes | archived |
| AES-HD | 5.30 | 1.78e-10 | yes | archived |
| AES-RD | 27.31 | 7.02e-11 | yes | archived |
| DPAv4 | 20.71 | 8.57e-10 | yes | archived |

[PASS] PI/EPI temperature sweep (full version): 18 values match the paper, 0 differ
[paper_results.fullversion_tables: 0 s, exit code 0]

================ summary ================
Table 7   matches the paper
Table 8   matches the paper
Table 9   matches the paper
Figure 1  matches the paper
Figure 2  matches the paper
Full-version tables matches the paper
tables and figures: results/paper_results
'created' timestamp seems very low; regarding as unix timestamp
'modified' timestamp seems very low; regarding as unix timestamp
[exit code 0, 20 s]
```

To see only the check lines:

```
python -m paper_results.reproduce archived --skip-oracle 2>/dev/null | grep -E '^\[(PASS|DIFF)\]'
```

```
$ python -m paper_results.reproduce archived --skip-oracle 2>/dev/null | grep -E '^\[(PASS|DIFF)\]'
[PASS] Table 7 (finite-suite certificates): 80 values match the paper, 0 differ
[PASS] Table 8 (key-byte recovery success): 28 values match the paper, 0 differ
[PASS] Table 9 (full-suite local-family certificates): 81 values match the paper, 0 differ
[PASS] Figure 1 (dimension sweep): 5 values match the paper, 0 differ
[PASS] Figure 2 (selected-center local-family certificates): 12 values match the paper, 0 differ
[PASS] Four-byte composition (full version): 1 values match the paper, 0 differ
[PASS] PI/EPI temperature sweep (full version): 18 values match the paper, 0 differ
[exit code 0, 4 s]
```

The tables and figures are written to `results/paper_results/`:

```
$ ls results/paper_results
bi_suite_dataset_summary.csv
figure1_dimension_sweep.csv
figure1_dimension_sweep.pdf
figure1_dimension_sweep.png
figure2_biloc_selected_centers.csv
figure2_biloc_selected_centers.pdf
figure2_biloc_selected_centers.png
figure2_biloc_selected_centers_full_attack_set.csv
figure2_calibrated_radius_points.csv
fullversion_epi_temperature.csv
fullversion_epi_temperature.md
fullversion_four_byte.csv
fullversion_four_byte.md
gaussian_oracle
table7_bi_suite.csv
table7_bi_suite.md
table8_key_recovery.csv
table8_key_recovery.md
table9_biloc_full_suite.csv
table9_biloc_full_suite.md
[exit code 0, 0 s]
```

```
$ cat results/paper_results/table7_bi_suite.md
### Table 7: finite-suite certificates (delta = 1e-06)
```

Dataset	d	M	n_te	p_obs	[p−, p+]	BI_obs	[BI−, BI+]	R_margin	EPI	counts
ASCAD d0	700	80	10,000	0.00630	[0.00260, 0.01247]	0.690	[0.000, 1.675]	2.58	−0.000	arc
hived
ASCAD d50	700	80	10,000	0.00970	[0.00485, 0.01701]	1.312	[0.311, 2.123]	1.26	0.239	arc
hived
ASCAD d100	700	80	10,000	0.00980	[0.00492, 0.01714]	1.327	[0.332, 2.134]	1.25	0.208	ar
chived
ASCAD random	1,400	11	100,000	0.01693	[0.01467, 0.01941]	2.116	[1.909, 2.313]	0.19	0.524
archived
AES-HD	1,250	80	25,000	0.00520	[0.00288, 0.00851]	0.413	[0.000, 1.124]	2.56	−0.006	arc
hived
AES-RD	3,500	80	12,500	0.13968	[0.12135, 0.15946]	5.160	[4.957, 5.351]	0.15	3.097	arch
ived
DPAv4	4,000	80	500	0.02400	[0.00201, 0.09144]	2.619	[0.000, 4.549]	3.36	2.251	archived
CHES-CTF-2025	3,000	11	100,000	0.00429	[0.00320, 0.00561]	0.135	[0.000, 0.521]	3.43	−0.1
05	archived
[exit code 0, 0 s]

$ cat results/paper_results/table8_key_recovery.md
Table 8: key-byte recovery success (%) over 100 shuffled orders

Benchmark	100	500	1,000	2,000	model	source
ASCAD d0	0	0	0	1	zaid_ascad_desync_0_feature_scaling_0_1_2	archived
ASCAD d50	49	100	100	100	zaid_ascad_desync_50_no_preprocessing_4	archived
ASCAD d100	67	100	100	100	zaid_ascad_desync_100_no_preprocessing_1	archived
ASCAD random	98	100	100	100	ascad_random_key_estranet_ckpt80	archived
AES-HD	1	4	13	30	zaid_aes_hd_no_preprocessing_7	archived
AES-RD	100	100	100	100	zaid_aes_rd_no_preprocessing_9	archived
DPAv4	100	100	---	---	zaid_dpav4_no_preprocessing_9	archived
CHES-CTF-2025	0	0	2	0	ches_ctf_2025_estranet_ckpt5	archived
[exit code 0, 0 s]

$ cat results/paper_results/table9_biloc_full_suite.md
Table 9: full-suite local-family certificate over M = 80 models (delta = 1e-06)

Benchmark	eps	S_loc	p_hat_loc	BI_loc_plus	Arch.	counts
AES-RD	0	1,746	0.140	5.36	CNN	archived
AES-RD	0.0001	1,756	0.140	5.37	CNN	archived
AES-RD	0.0003	1,767	0.141	5.37	CNN	archived
AES-RD	0.001	1,812	0.145	5.41	CNN	archived
AES-RD	0.01	11,737	0.939	7.93	MLP	archived
ASCAD d0	0	63	0.006	1.71	CNN	archived
ASCAD d0	0.0001	264	0.026	3.28	MLP	archived
ASCAD d0	0.0003	1,694	0.169	5.64	CNN	archived
ASCAD d0	0.001	10,000	1.000	8.00	MLP	archived
ASCAD d0	0.01	10,000	1.000	8.00	MLP	archived
ASCAD d50	0	97	0.010	2.15	CNN	archived
ASCAD d50	0.0001	112	0.011	2.30	CNN	archived
ASCAD d50	0.0003	141	0.014	2.56	CNN	archived
ASCAD d50	0.001	815	0.082	4.68	CNN	archived
ASCAD d50	0.01	10,000	1.000	8.00	CNN	archived
ASCAD d100	0	98	0.010	2.16	CNN	archived
ASCAD d100	0.0001	109	0.011	2.28	CNN	archived
ASCAD d100	0.0003	233	0.023	3.13	MLP	archived
ASCAD d100	0.001	4,936	0.494	7.07	MLP	archived
ASCAD d100	0.01	9,934	0.993	8.00	MLP	archived
[exit code 0, 0 s]

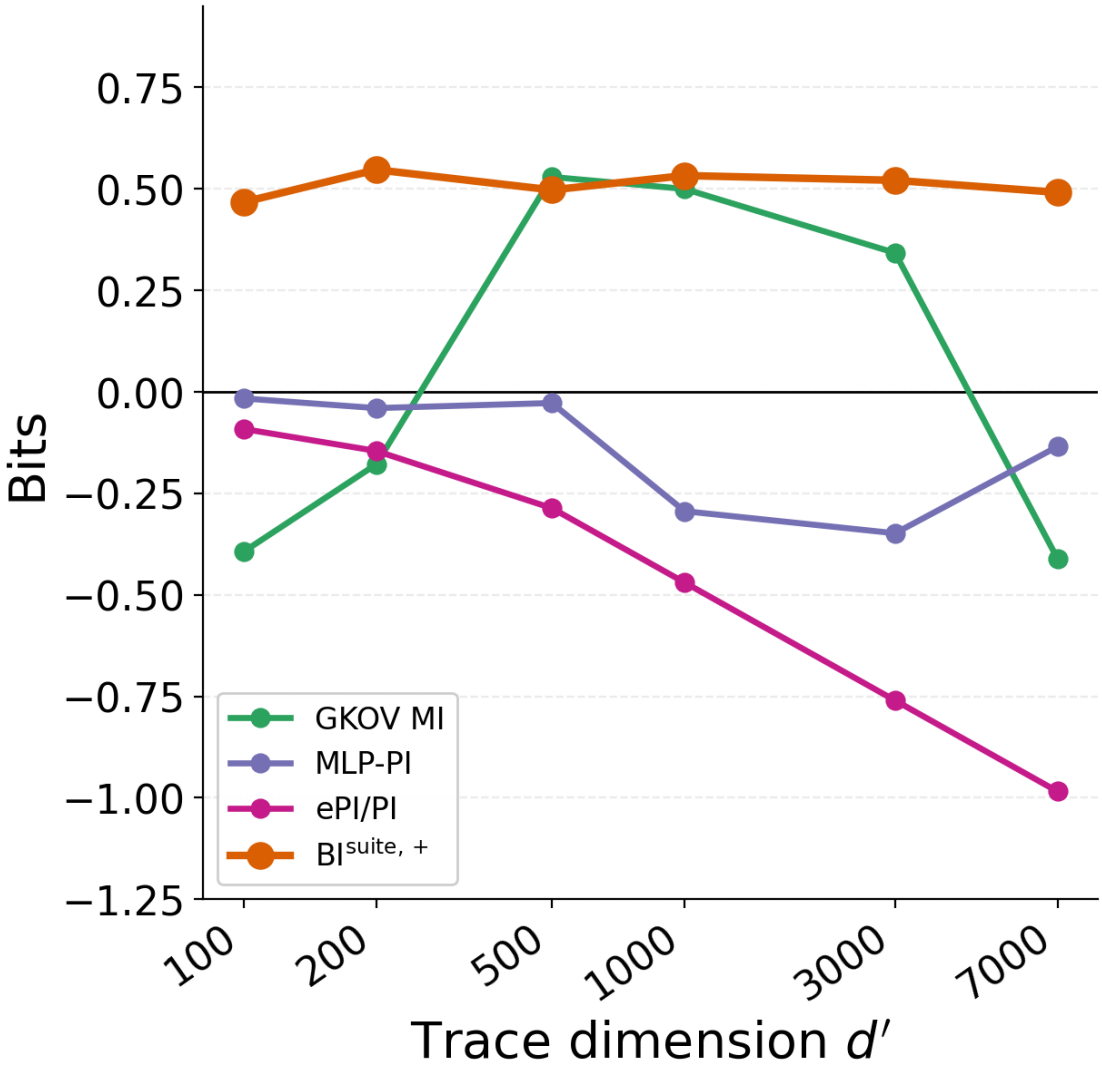


Figure 1: Figure 1, as rebuilt by the code.

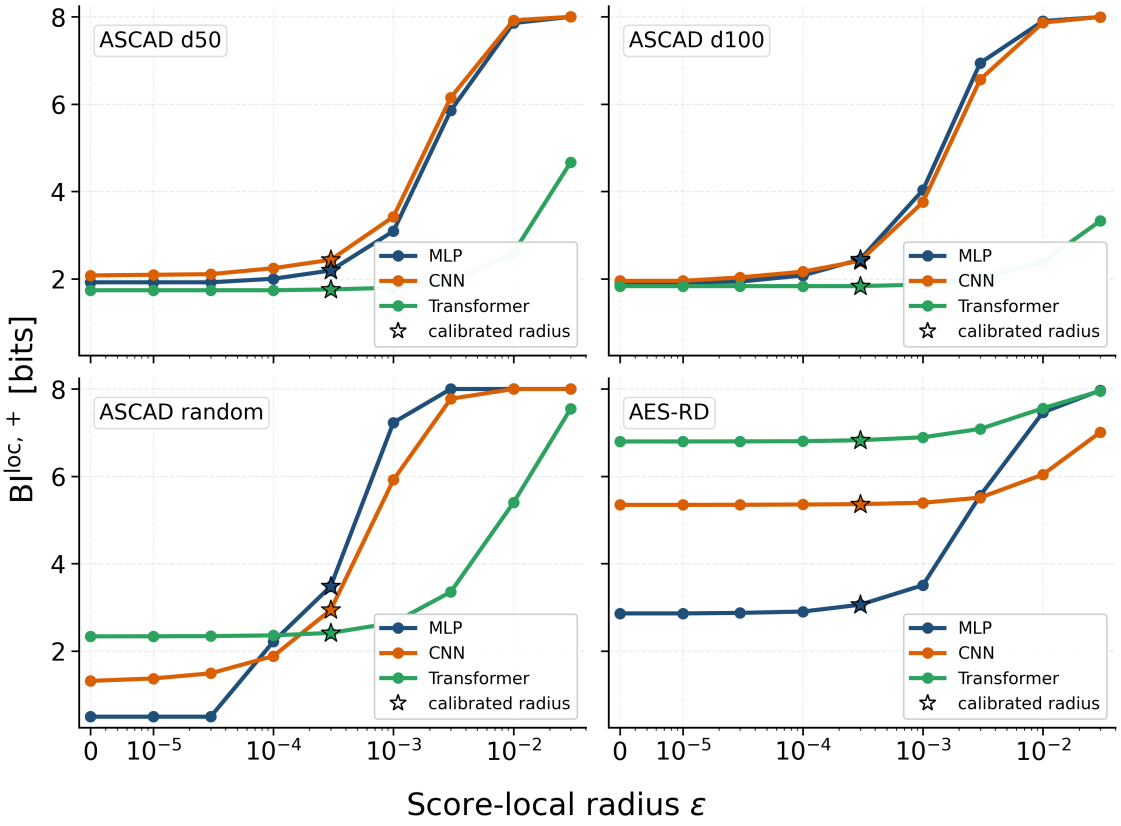


Figure 2: Figure 2, as rebuilt by the code.

What to look for. Eight `[PASS]` lines, and `matches the paper` for every result. In the Table 7 file, `p_obs` is the success rate of the best model in the suite, `p_plus` is the guaranteed upper limit on that rate, and `BI_plus` is the same limit in bits. The last column says whether a row used saved or recomputed counts.

Level B: recompute from the TCHES20 traces and models

Download the traces and the 480 trained models (about a minute):

```
paper_results/fetch_tches20.sh
```

```
$ paper_results/fetch_tches20.sh
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/ASCAD_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/AES_HD_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/AES_RD_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/DPAv4_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/models/pretrained_models.zip
TCHES20 benchmark ready in datasets/raw/tches20/TCHES20V3_CNN_SCA (480 pretrained models)
Cloning into 'datasets/raw/tches20/TCHES20V3_CNN_SCA'...
[exit code 0, 14 s]
```

Run the level. It tests all 80 models of each trace set, counts their correct guesses, and rebuilds the tables from those counts:

```
python -m paper_results.reproduce tches20
```

```
$ python -m paper_results.reproduce tches20
[... earlier output omitted; key lines kept ...]
$ python -m bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split --dataset ascad_desync_0 --split atta
    ck --models-dir datasets/raw/tches20/TCHES20V3_CNN_SCA/models/pretrained_models/models --dat …
```

```
{'dataset': 'ascad_desync_0', 'split': 'attack', 'scope_id': 'ascad_desync_0_tches20_m80_attack_fixed_split'
    , 'M': 80, 'n_te': 10000, 'p_obs': 0.0063, 'p_minus': 0.0025954814754893776, 'p_plus': 0.012 …
[bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split: 15 s, exit code 0]
$ python -m bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split --dataset ascad_desync_50 --split att
    ack --models-dir datasets/raw/tches20/TCHES20V3_CNN_SCA/models/pretrained_models/models --da …
{'dataset': 'ascad_desync_50', 'split': 'attack', 'scope_id': 'ascad_desync_50_tches20_m80_attack_fixed_spli
    t', 'M': 80, 'n_te': 10000, 'p_obs': 0.0097, 'p_minus': 0.0048463177805206144, 'p_plus': 0.0 …
[bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split: 36 s, exit code 0]
$ python -m bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split --dataset ascad_desync_100 --split at
    tack --models-dir datasets/raw/tches20/TCHES20V3_CNN_SCA/models/pretrained_models/models --d …
{'dataset': 'ascad_desync_100', 'split': 'attack', 'scope_id': 'ascad_desync_100_tches20_m80_attack_fixed_sp
    lit', 'M': 80, 'n_te': 10000, 'p_obs': 0.0098, 'p_minus': 0.004915831105724693, 'p_plus': 0. …
[bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split: 40 s, exit code 0]
$ python -m bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split --dataset aes_hd --split attack --mod
    els-dir datasets/raw/tches20/TCHES20V3_CNN_SCA/models/pretrained_models/models --dataset-dir …
{'dataset': 'aes_hd', 'split': 'attack', 'scope_id': 'aes_hd_tches20_m80_attack_fixed_split', 'M': 80, 'n_te
    ': 25000, 'p_obs': 0.0052, 'p_minus': 0.0028794913844254534, 'p_plus': 0.008513928525316905, …
[... 116 lines omitted ...]
$ python -m paper_results.table7_bi_suite --recomputed-dir results/paper_results/recomputed --out-dir result
    s/paper_results
ascad_desync_0: 80/80 recomputed per-model counts equal the archived counts
ascad_desync_50: 80/80 recomputed per-model counts equal the archived counts
ascad_desync_100: 80/80 recomputed per-model counts equal the archived counts
aes_hd: 80/80 recomputed per-model counts equal the archived counts
aes_rd: 80/80 recomputed per-model counts equal the archived counts
dpav4: 80/80 recomputed per-model counts equal the archived counts
| ASCAD d0 | 700 | 80 | 10,000 | 0.00630 | [0.00260, 0.01247] | 0.690 | [0.000, 1.675] | 2.58 | -0.000 | rec
    omputed |
| ASCAD d50 | 700 | 80 | 10,000 | 0.00970 | [0.00485, 0.01701] | 1.312 | [0.311, 2.123] | 1.26 | 0.239 | rec
    omputed |
| ASCAD d100 | 700 | 80 | 10,000 | 0.00980 | [0.00492, 0.01714] | 1.327 | [0.332, 2.134] | 1.25 | 0.208 | re
    computed |
| AES-HD | 1,250 | 80 | 25,000 | 0.00520 | [0.00288, 0.00851] | 0.413 | [0.000, 1.124] | 2.56 | -0.006 | rec
    omputed |
| AES-RD | 3,500 | 80 | 12,500 | 0.13968 | [0.12135, 0.15946] | 5.160 | [4.957, 5.351] | 0.15 | 3.097 | reco
    mputed |
| DPAv4 | 4,000 | 80 | 500 | 0.02400 | [0.00201, 0.09144] | 2.619 | [0.000, 4.549] | 3.36 | 2.251 | recomput
    ed |
[PASS] Table 7 (finite-suite certificates): 80 values match the paper, 0 differ
results/paper_results/recomputed/tches20_full_suite_loc/aes_rd_full_suite_bi_loc_summary.csv
results/paper_results/recomputed/tches20_full_suite_loc/ascad_desync_0_full_suite_bi_loc_summary.csv
results/paper_results/recomputed/tches20_full_suite_loc/ascad_desync_50_full_suite_bi_loc_summary.csv
results/paper_results/recomputed/tches20_full_suite_loc/ascad_desync_100_full_suite_bi_loc_summary.csv
$ python -m bi_scope_certificates.profile_epsilon_calibration --dataset ascad_desync_50 --public-only --tche
    s-root datasets/raw/tches20/TCHES20V3_CNN_SCA --output-dir results/paper_results/recomputed/ …
$ python -m bi_scope_certificates.profile_epsilon_calibration --dataset ascad_desync_100 --public-only --tch
    es-root datasets/raw/tches20/TCHES20V3_CNN_SCA --output-dir results/paper_results/recomputed …
WARNING:tensorflow:6 out of the last 6 calls to <function Model.make_predict_function.<locals>.predict_funct
    ion at 0x7274ac556160> triggered tf.function retracing. Tracing is expensive and the excessi …

100%|██████████| 10000/10000 [00:02<00:00, 4442.42it/s]

100%|██████████| 10000/10000 [00:02<00:00, 4466.87it/s]

100%|██████████| 9963/10000 [00:02<00:00, 4525.13it/s]
100%|██████████| 10000/10000 [00:02<00:00, 4507.64it/s]

100%|██████████| 24954/25000 [00:10<00:00, 2305.33it/s]
100%|██████████| 25000/25000 [00:10<00:00, 2297.90it/s]

100%|██████████| 24904/25000 [00:05<00:00, 4477.57it/s]
100%|██████████| 25000/25000 [00:05<00:00, 4507.33it/s]

100%|██████████| 500/500 [00:00<00:00, 2370.30it/s]
'created' timestamp seems very low; regarding as unix timestamp
'modified' timestamp seems very low; regarding as unix timestamp
[exit code 0, 6 min 49 s]
```

```
$ ls results/paper_results/recomputed/tches20_suite && \
    head -5 results/paper_results/recomputed/tches20_suite/ascad_desync_0_attack_success.csv
```

```
aes_hd_attack_success.csv
aes_hd_attack_summary.csv
aes_rd_attack_success.csv
aes_rd_attack_summary.csv
ascad_desync_0_attack_success.csv
ascad_desync_0_attack_summary.csv
ascad_desync_100_attack_success.csv
ascad_desync_100_attack_summary.csv
ascad_desync_50_attack_success.csv
ascad_desync_50_attack_summary.csv
dpav4_attack_success.csv
dpav4_attack_summary.csv
dataset,split,model_id,model_path,n_te,successes,success_rate,status,error,scope_id
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_0,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_0.hdf5,10000,42,0.0 …
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_1,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_1.hdf5,10000,38,0.0 …
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_2,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_2.hdf5,10000,39,0.0 …
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_3,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_3.hdf5,10000,41,0.0 …
[exit code 0, 0 s]
```

What to look for. `80/80 recomputed per-model counts equal the archived counts` for each of the six trace sets. A count is the number of traces where the model's top guess is right, so it comes out the same on any machine.

Shortcuts: `--quick` tests only the six best models (two minutes). `--datasets` limits the run to some trace sets. A stopped run continues where it left off.

Level C: recompute with the Transformer models

Get the Transformer source code:

```
paper_results/fetch_estranet.sh
```

```
$ paper_results/fetch_estranet.sh
EstraNet sources ready in external/EstraNet
Cloning into 'external/EstraNet'...
[exit code 0, 1 s]
```

Download, check and unpack the provided models (308 MB):

```
python -m paper_results.fetch_checkpoints
ls checkpoints/estranet checkpoints/keras_ascad_random_key
```

```
$ python -m paper_results.fetch_checkpoints && \
    ls checkpoints/estranet checkpoints/keras_ascad_random_key
downloading https://github.com/K1015/PAC-Certification-of-Multivariate-Side-Channel-Traces/releases/download
    /v1.0/bi_certification_checkpoints.tar.gz
bi_certification_checkpoints.tar.gz: checksum verified
unpacked and verified 162 files in checkpoints
checkpoints/estranet:
aes_rd_fstd_pool5_lr5e4
ascadf_desync100
ascadf_desync50
ascadr
ches25_lx
ches25_lx_w100
ches25_lx_w1000
ches25_lx_w200
ches25_lx_w3000
ches25_lx_w500

checkpoints/keras_ascad_random_key:
ascad_random_key_cnn_seed0_best.hdf5
```

```
ascad_random_key_mlp_seed0_best.hdf5
[exit code 0, 8 s]
```

Download the ASCAD random-key traces:

```
python -m paper_results.fetch_ascad_random_key
```

```
$ python -m paper_results.fetch_ascad_random_key
downloading https://www.data.gouv.fr/api/1/datasets/r/b4ace767-c2a4-4db4-8e01-4527b5b91f00
datasets/raw/ascad_random_key/ASCAD.h5: checksum verified
[exit code 0, 18 s]
```

Download `CHES_Challenge.h5` by hand from https://pace-tl.gitbook.io/ches-challenge-2025. Put it in the folder `datasets/raw/ches_ctf_2025/` and check it:

```
$ ls -l datasets/raw/ches_ctf_2025 && \
    sha256sum datasets/raw/ches_ctf_2025/CHES_Challenge.h5
total 4120908
-rw-r--r-- 2 1000 1000 4219804480 Sep 30 15:34 CHES_Challenge.h5
132ae2e9a8213c983bf3b63449e9572d5d71d3b376a75b236415d4a728b9379f  datasets/raw/ches_ctf_2025/CHES_Challenge.
    h5
[exit code 0, 11 s]
```

Run the two Table 7 suites. Each one tests eleven saved models on 100,000 traces:

```
python -m paper_results.reproduce estranet --suites ascad_random_key ches_ctf_2025
```

```
$ python -m paper_results.reproduce estranet --suites ascad_random_key ches_ctf_2025
[... earlier output omitted ...]
 ascad_desync_50          CNN           zaid_ascad_desync_50_no_preprocessing_3           0.0003  8000
       97        2.436
 ascad_desync_50          MLP        noConv1_ascad_desync_50_no_preprocessing_0           0.0003  8000
       77        2.192
 ascad_desync_50  Transformer                  ascad_desync_50_estranet_ckpt60           0.0003  8000
       50        1.759
ascad_random_key          CNN                       ascad_random_key_cnn_seed0           0.0003 80000
     2112        2.941
ascad_random_key          MLP                       ascad_random_key_mlp_seed0           0.0003 80000
     3145        3.482
ascad_random_key  Transformer                 ascad_random_key_estranet_ckpt80           0.0003 80000
     1428        2.416

results/paper_results/figure2_biloc_selected_centers.pdf
[PASS] Figure 2 (selected-center local-family certificates): 12 values match the paper, 0 differ
[paper_results.figure2_biloc_selected_centers: 2 s, exit code 0]

[... 31 lines omitted ...]
$ python -m paper_results.fullversion_tables --recomputed-dir results/paper_results/recomputed --out-dir res
    ults/paper_results
[PASS] Four-byte composition (full version): 1 values match the paper, 0 differ
[PASS] PI/EPI temperature sweep (full version): 18 values match the paper, 0 differ
Figure 1  matches the paper
Figure 2  matches the paper
Full-version tables matches the paper
tables and figures: results/paper_results
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
```

```
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
'created' timestamp seems very low; regarding as unix timestamp
'modified' timestamp seems very low; regarding as unix timestamp
[exit code 0, 16 min 12 s]
```

Optional: the five extra suites behind the BI line of Figure 1 (models trained on shorter and longer traces), and the twelve models of Figure 2:

```
python -m paper_results.reproduce estranet --suites ches_windows
```

```
$ python -m paper_results.reproduce estranet --suites ches_windows
[... earlier output omitted ...]

$ python -m paper_results.figure1_dimension_sweep --recomputed-dir results/paper_results/recomputed --out-di
    r results/paper_results
window 100: 9/11 recomputed per-checkpoint counts equal the archived counts (largest difference 1 traces)
window 200: 9/11 recomputed per-checkpoint counts equal the archived counts (largest difference 1 traces)
window 500: 9/11 recomputed per-checkpoint counts equal the archived counts (largest difference 1 traces)
window 1000: 8/11 recomputed per-checkpoint counts equal the archived counts (largest difference 1 traces)
window 3000: 11/11 recomputed per-checkpoint counts equal the archived counts (largest difference 0 traces)
window 7000: 11/11 recomputed per-checkpoint counts equal the archived counts (largest difference 0 traces)
dimension  BI_suite,+      GKOV MI       MLP-PI       ePI/PI
      100       0.467      -0.394       -0.016       -0.091
      200       0.544      -0.178       -0.040       -0.145
[... 81 lines omitted ...]
[PASS] Figure 1 (dimension sweep): 5 values match the paper, 0 differ
$ python -m paper_results.figure2_biloc_selected_centers --recomputed-dir results/paper_results/recomputed -
    -out-dir results/paper_results
[PASS] Figure 2 (selected-center local-family certificates): 12 values match the paper, 0 differ
$ python -m paper_results.fullversion_tables --recomputed-dir results/paper_results/recomputed --out-dir res
    ults/paper_results
[PASS] Four-byte composition (full version): 1 values match the paper, 0 differ
[PASS] PI/EPI temperature sweep (full version): 18 values match the paper, 0 differ
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
'created' timestamp seems very low; regarding as unix timestamp
'modified' timestamp seems very low; regarding as unix timestamp
[exit code 0, 29 min 42 s]
```

```
python -m bi_scope_certificates.prepare_aesrd_estranet_h5
for ds in ascad_random_key ascad_desync_50 ascad_desync_100 aes_rd; do
  python -m bi_scope_certificates.profile_epsilon_calibration --dataset $ds \
    --output-dir results/paper_results/recomputed/selected_centers/$ds
done
python -m paper_results.reproduce report
```

```
$ python -m bi_scope_certificates.prepare_aesrd_estranet_h5 && \
    for ds in ascad_random_key ascad_desync_50 ascad_desync_100 aes_rd; do python -m bi_scope_certificates.p
    rofile_epsilon_calibration --dataset $ds --output-dir results/paper_results/recomputed/selected_centers/
    $ds; done && \
    python -m paper_results.reproduce report
[... earlier output omitted ...]
ascad_random_key          MLP                        ascad_random_key_mlp_seed0           0.0003 80000
     3145        3.482
ascad_random_key  Transformer                  ascad_random_key_estranet_ckpt80           0.0003 80000
     1428        2.416

results/paper_results/figure2_biloc_selected_centers.pdf
[PASS] Figure 2 (selected-center local-family certificates): 12 values match the paper, 0 differ
[paper_results.figure2_biloc_selected_centers: 2 s, exit code 0]

$ python -m paper_results.fullversion_tables --recomputed-dir results/paper_results/recomputed --out-dir res
    ults/paper_results
### Full version: raw-ASCAD four-byte composition

| Traces | Printed | Printed rate | Archived final models (1,000 campaigns) | Rerun | Rerun rate | Rerun med
    ian rank |
[... 31 lines omitted ...]
[PASS] Four-byte composition (full version): 1 values match the paper, 0 differ
[PASS] PI/EPI temperature sweep (full version): 18 values match the paper, 0 differ
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
'created' timestamp seems very low; regarding as unix timestamp
'modified' timestamp seems very low; regarding as unix timestamp
[exit code 0, 16 min 31 s]
```

What to look for. The best model of each suite is the same as in the paper (ASCAD random key: model 80 with 1,693 correct guesses out of 100,000; CHES-CTF-2025: model 5 with 429), and every check ends with `[PASS]`. For Figure 2, the output prints the chosen safety margin, `0.0003`, as in the paper.

Level D: the two results quoted from the long version

```
python -m paper_results.reproduce fullversion
```

```
$ python -m paper_results.reproduce fullversion
[... earlier output omitted ...]
          aes_rd          MLP                 noConv1_aes_rd_no_preprocessing_2           0.0003 10000
      224        3.061
          aes_rd  Transformer aes_rd_estranet_fstd_pool5_lr5e4_6977529_2_ckpt11           0.0003 20000
     8447        6.828
ascad_desync_100          CNN          zaid_ascad_desync_100_no_preprocessing_5           0.0003  8000
       95        2.413
ascad_desync_100          MLP       noConv1_ascad_desync_100_no_preprocessing_6           0.0003  8000
```

```
       98        2.447
ascad_desync_100  Transformer                  ascad_desync_100_estranet_ckpt40           0.0003  8000
       54        1.834
 ascad_desync_50          CNN           zaid_ascad_desync_50_no_preprocessing_3           0.0003  8000
       97        2.436
 ascad_desync_50          MLP        noConv1_ascad_desync_50_no_preprocessing_0           0.0003  8000
       77        2.192
 ascad_desync_50  Transformer                   ascad_desync_50_estranet_ckpt60           0.0003  8000
       50        1.759
ascad_random_key          CNN                        ascad_random_key_cnn_seed0           0.0003 80000
     2112        2.941
ascad_random_key          MLP                        ascad_random_key_mlp_seed0           0.0003 80000
     3145        3.482
ascad_random_key  Transformer                  ascad_random_key_estranet_ckpt80           0.0003 80000
     1428        2.416
[... 40 lines omitted ...]
[PASS] Figure 2 (selected-center local-family certificates): 12 values match the paper, 0 differ
$ python -m paper_results.fullversion_tables --recomputed-dir results/paper_results/recomputed --out-dir res
    ults/paper_results
[PASS] Four-byte composition (full version): 6 values match the paper, 0 differ
| ASCAD d0 | 4.23 | 1.46e-10 | yes | recomputed |
| ASCAD d50 | 23.07 | 5.07e-10 | yes | recomputed |
| ASCAD d100 | 22.48 | 4.39e-10 | yes | recomputed |
| AES-HD | 5.30 | 4.24e-10 | yes | recomputed |
| AES-RD | 27.31 | 7.37e-11 | yes | recomputed |
| DPAv4 | 20.71 | 1.70e-09 | yes | recomputed |
[PASS] PI/EPI temperature sweep (full version): 18 values match the paper, 0 differ
100%|██████████| 10000/10000 [00:02<00:00, 4798.73it/s]

100%|██████████| 10000/10000 [00:02<00:00, 4802.12it/s]

100%|██████████| 10000/10000 [00:02<00:00, 4802.33it/s]

100%|██████████| 24985/25000 [00:10<00:00, 2448.31it/s]
100%|██████████| 25000/25000 [00:10<00:00, 2432.23it/s]

100%|██████████| 25000/25000 [00:05<00:00, 4844.52it/s]
WARNING:tensorflow:5 out of the last 9 calls to <function Model.make_predict_function.<locals>.predict_funct
    ion at 0x760cd9625a80> triggered tf.function retracing. Tracing is expensive and the excessi …
WARNING:tensorflow:6 out of the last 11 calls to <function Model.make_predict_function.<locals>.predict_func
    tion at 0x760cd9624720> triggered tf.function retracing. Tracing is expensive and the excess …

100%|██████████| 500/500 [00:00<00:00, 2508.83it/s]
WARNING:tensorflow:5 out of the last 5 calls to <function Model.make_predict_function.<locals>.predict_funct
    ion at 0x7ca1119be8e0> triggered tf.function retracing. Tracing is expensive and the excessi …
WARNING:tensorflow:6 out of the last 6 calls to <function Model.make_predict_function.<locals>.predict_funct
    ion at 0x7ca1119bee80> triggered tf.function retracing. Tracing is expensive and the excessi …
'created' timestamp seems very low; regarding as unix timestamp
'modified' timestamp seems very low; regarding as unix timestamp
[exit code 0, 10 min 5 s]
```

What to look for. First, a comparison of two information measures on six trace sets. Then the raw ASCAD traces are downloaded and checked, points of interest are found, eight small models are trained (one per key byte), and four of them are combined to recover four key bytes at once. The check ends with `[PASS]` for both results. The four-byte check tests that the four key bytes are recovered from 500 traces on.

`python -m paper_results.reproduce report` rebuilds the tables from whatever has been recomputed so far.